Fiscal Year:	FY 2019	**Task Last Updated:**	FY 05/15/2019
PI Name:	Kiss, John Z Ph.D.		
Project Title:	Novel Explorations Into The Interactions Between Light and Gravity Sensing in Plants		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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City:	Greensboro	**State:**	NC
Zip Code:	27402-6170	**Congressional District:**	13
Comments:	NOTE: PI moved to Florida Institute of Technology from University of North Carolina-Greensboro in May 2024 (Ed., 2/6/25). NOTE: PI moved to University of North Carolina-Greensboro from University of Mississippi in July 2016 (Ed., 6/13/17) NOTE: PI moved to U Miss from Miami (Ohio) University in September 2012 (Ed., 8/10/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	08/10/2017	**End Date:**	07/31/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 7/31/2020 per NSSC information (Ed., 5/15/19)		
Key Personnel Changes/Previous PI:	May 2019 report: New post-doc: Dr. Tatsiana Shymanovich.		
COI Name (Institution):			
Grant/Contract No.:	80NSSC17K0546		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	[ED. NOTE 12/11/2017: Continuation of "Novel Explorations Into the Interactions Between Light and Gravity Sensing in Plants--NNX12AO65G," per PI move from University of Mississippi to University of North Carolina at Greensboro. See that project for previous reports] The major goals of this project are to determine how gravity and light responses influence each other in plants and to better understand the cellular signaling mechanisms involved in plant tropisms. This proposed project builds on our previous spaceflight experiments on the International Space Station (ISS) with the European Modular Cultivation System (EMCS) using the model plant Arabidopsis. In this new proposed project, we plan to confirm and extend our discovery during the recent ISS experiment of a novel red-light-based positive phototropic response in plant shoots. Our hypothesis is that positive red-light-sensing, which was known in older plant lineages, is masked by normal 1-g conditions in more recently evolved lineages. We also propose to investigate fundamental interactions among red and blue light signaling pathways and the gravity sensing mechanisms. The ISS is the only laboratory facility where it is possible to reliably and simultaneously vary light and gravity stimuli for the proposed experiments. These experiments can be accomplished with telemetric science and therefore fit with the extremely limited down mass capabilities of current spaceflight opportunities. The EMCS is an automated facility and can be operated largely with telemetry. Experimental containers will launched via Space-X, and during an experimental run, images of seedling growth responses can be downlinked to Earth. A significant advantage of this proposed project is that we will use existing flight hardware that has been flown successfully and allowed for growth of Arabidopsis seedlings in the EMCS. This project is also relevant to the recent emphasis at NASA to study plant growth and development at fractional g-levels such as those found on the moon and Mars. Improved knowledge of the basic mechanistic processes that will be the focus of this project is vital to develop ways to use plants in extraterrestrial bioregenerative life support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Studying the effects of gravity and light on basic cellular mechanisms in plants, this project is relevant to the emphasis on using Arabidopsis thaliana as a model system in developmental biology. Since plants will be a necessary part of bioregenerative life support needed to send humans to Mars and beyond, the knowledge obtained from our spaceflight experiments will be critical for developing ways to effectively use plants in the bioregenerative life support systems. These life support systems also can be used in ground-based Earth applications such as the design of large-scale greenhouses to support crop plants and agriculture. In addition, improved knowledge of light and gravity systems in plants will, in the long term, allow us to develop better crop plants for use on Earth.
Task Progress:	We utilized the condition of microgravity on board the International Space Station (ISS) to identify genes involved in the interaction between light responses (phototropism) and gravity responses (gravitropism) in Arabidopsis thaliana seedlings. Seedlings were grown on the ISS, and RNA was extracted from 7 samples (pools of 10-15 plants) grown in microgravity (µg) or Earth gravity conditions (1.0 g). Transcriptomic analysis of differential gene expression was performed using the HISAT2-Stringtie-DESeq2 RNASeq pipeline. Differentially expressed genes were further characterized by using Pathway Analysis as well as enrichment for Gene Ontology classifications. 296 genes were found to be significantly differentially expressed between microgravity when compared to 1-g controls ($p < 0.05$). In addition, Pathway Analysis identified 8 molecular pathways that are significantly affected by reduced gravity conditions. Specifically, light-associated pathways (e.g., photosynthesis-antenna proteins, photosynthesis, porphyrin and chlorophyll metabolism) show significant downregulation. Arabidopsis thaliana seedlings grown in conditions of microgravity show significant alterations to gene expression when compared to the 1-g control. Understanding the interactions between these two essential tropistic responses not only furthers our fundamental understanding, but may also help to efficiently grow plants during long-range space missions. In addition, this study is the first to use RNAseq tools to study the interactions between tropisms in a spaceflight study.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
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Abstracts for Journals and Proceedings	Vandenbrink JP, Edelmann RE, Herranz R, Medine FJ, Kiss JZ. "Analysis of genes associated with a novel blue-light phototropic response in plants utilizing RNA-Seq." 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. Abstract Book, p. 33. , Nov-2018
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